

The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.6 (Automated Adversarial Testing of the Guarded EBU Ledger)

This is an adversarial-search study, NOT an economy. No prices, ownership, or transferable EBU are introduced. It is a falsification stage over the frozen v2.4.0 physics and V2.5 guarded ledger, asking one question:

Can an automated actor or coalition find action sequences that earn positive cumulative guarded EBU while causing persistent physical harm?

Result. The hand-built red-team fixture found no guarded failure and it rediscovers a naive-ledger exploit (positive control passed). But a **corrected, properly paired randomized study found a CONFIRMED profitable persistent-harm exploit against the guarded ledger**: on one of twelve random layouts (seed 0) the guarded greedy adversary earns positive net EBU (+260) yet drives every regenerative source below its Allee threshold; with a 20-tick no-action persistence tail all five sources die and viability reaches 0%. Guarded EBU therefore does **not** guarantee incentive-alignment on every topology. The ledger is reported unchanged (not patched); a corrected variant is left to the next stage.

1. Objective and Non-Goals

V2.5 showed a hand-written attack list could not game the guarded ledger. A finite hand list is weak evidence. V2.6 replaces it with an automated, deterministic search. No economic features are added; the frozen files (energy_balance, ebu_v22..v25) are unchanged and re-verified. All V2.6 code lives in new files so earlier results stay reproducible.

2. Method

2.1 Forced-action harness (reuses the frozen accounting)

A tick harness applies an arbitrary list of feasibility-clipped actor actions through the exact V2.5 accounting (the same `_action_effect`), so guarded/naive semantics are identical to V2.5 and the physical trajectory under physics selection is byte-identical to the frozen baseline (tested). Every tick also checks the conservation identity $dX = S + G - D - \text{Lambda} - \text{loss} - \text{spill}$.

2.2 Deterministic beam search

A beam search plans multi-tick, multi-actor sequences. Per tick each coalition actor chooses from a menu `{rest} + transfers` to each admissible neighbour at exactly **two** transfer quantities (0.5 and 1.0 of `q_max`); the joint action is the product over the coalition; the search uses **depth 10, width 40**, and the exploit test scans only the **final-depth beam states**. The action space thus spans neighbouring transfers, small splits, repeated/multi-tick actions, relays, cyclic/back-and-forth moves, coordinated multi-actor moves, and harvesting near or below regenerative reserves. The search is deterministic and consumes no randomness (tested), so it cannot peek at future events.

2.3 Two separated objectives

(I) an **automated adversary** maximising its own cumulative guarded EBU; (II) a **red-team** maximising cumulative physical harm subject to net EBU $\geq$ a positive threshold. These are kept apart throughout.

2.4 Exploit definition (fixed before the final runs)

An exploit requires ALL of:

1. coalition cumulative net guarded EBU > 0 (profitable)
2. physical condition persistently worse than a PAIRED no-action baseline (coalition rests, natural dynamics)
3. the deterioration holds for a declared persistence window (every tail tick, not a one-tick blip)
4. harm measured on PHYSICAL variables (threshold-aware burden `B_R`), never on EBU itself.

Reported physical harm variables: final and post-shock viability, sustained recovery, unmet demand, dead regenerative sources, Allee-threshold crossings, source stock, `B_R`, transport dissipation. Reported EBU variables: issued credit, issued debit, net EBU, balances, concentration, plus search runtime and explored states.

3. Tests

48 tests pass: the 33 prior tests (verified inline) plus 15 new V2.6 behavior tests — observational trajectory exactly unchanged; conservation during multi-action sequences; search reproducibility; genuinely distinct random layouts; all 12 study masks pairwise distinct; the baseline/forced-tick harness is side-effect-free on the caller's grid (the fix for the Study C pairing bug); a reproducible seed-0 guarded-exploit regression fixture; rediscovery of a naive exploit; coalition totals = issued credit - debit; a lossless restoring multi-actor cycle earns no positive EBU; splitting cannot increase reward on the same trajectory; the exploit classifier

requires BOTH positive net AND persistent harm; the search uses no future randomness; physics-only identical to the frozen baseline.

4. Results

4.1 (A) Five policies from an identical initial state

Red-team fixture: a 3x3 world, an Allee source at the centre, eight deficient rim consumers, a two-actor coalition, 10 ticks. (This harsh, under-served world holds viability near 11% for every reasoned policy; it is a search fixture, not a homeostasis demo.)

Policy	viable %	B_R	dead	src stock	coalition net EBU
1 physics / no EBU	11.1	77.8	0	8.6	0.00
2 naive greedy adversary	11.1	88.9	0	12.5	+15.13
3 guarded greedy adversary	11.1	71.1	0	9.0	+21.21
4 random adversary	22.2	206.2	1	0.0	-126.69
5 automated guarded (beam)	11.1	78.2	0	8.9	+13.82

Observation: the random adversary is the only one that clearly worsens the field (B_R 206, a dead source) and it loses EBU (-126.69); the beam explored 22,743 states.

4.2 (B) Red-team search for profitable persistent harm

Ledger	verdict	net EBU	persistent harm	mean tail-harm (B_R)	states
naive (positive control)	EXPLOIT FOUND	+15.35	yes	+102.91	3,809
guarded	no exploit found	+0.91	no	-5.47	10,971

Experimental result: the search rediscovers a naive exploit — the coalition harvests the Allee source below its threshold to serve deficient consumers, earning positive naive credit (plain-burden improvement) at zero naive cost while the source dies; harm persists through the whole tail. Under the guarded ledger the same search finds no profitable persistent harm: the best net-positive sequence it finds actually leaves the field better than no action (mean tail-harm -5.47).

Interpretation: the naive positive control passed (the search reproduces a known ledger weakness). Its failure to break the guarded ledger on this ONE small fixture is therefore meaningful but narrow — and is contradicted by the randomized study below.

4.3 (C) Randomized-layout topology study (corrected, properly paired)

Correction. An earlier version of this study was mis-paired: the no-action baseline evaluation mutated the shared grid, so the guarded run began from the evolved state rather than the initial layout. The harness is now side-effect-free and baseline vs guarded run on separate but identical worlds per seed (both verified by tests). The numbers below are from the corrected run. Guarded *greedy* adversary vs a paired no-action baseline, 12 genuinely distinct random 5x5 Allee layouts, 30-tick attack + 20-tick persistence tail. Harm@30 = B_R(attack@30) - B_R(no-action@30); positive means worse than doing nothing.

Metric	mean	sd	min	max
coalition net guarded EBU	156.79	58.82	89.17	263.54
final viability %	80.33	23.29	20.00	100.00
harm@30: B_R(attack) - B_R(no-action)	-241.09	89.96	-302.82	+40.74

Experimental result: on average the greedy guarded adversary is strongly net-helpful (mean harm -241, i.e. much better than no action). But 1 of 12 layouts is positive-reward harmful, and applying the FULL persistence definition confirms it is a genuine exploit.

Confirmed guarded exploit (seed 0): net EBU +260.13; harm@30 +40.74; 4 of 5 sources already dead at tick 30; with the 20-tick no-action persistence tail the sequence satisfies the persistent-harm predicate (mean tail-harm +309.93); all 5 sources eventually die and viability reaches 0%. This is a profitable trajectory that leaves the world persistently, irreversibly worse than doing nothing.

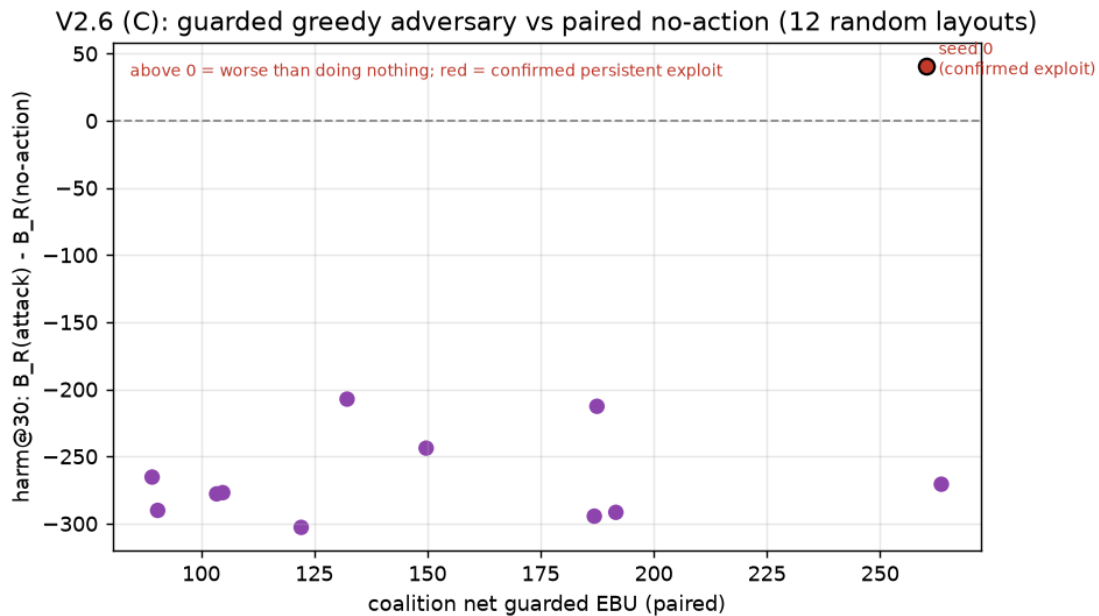


Figure 1. Corrected, paired study. Eleven layouts sit below zero (guarded adversary better than no action); the single red point (seed 0) is a confirmed profitable persistent-harm exploit (positive net EBU, worse than no action, all sources die under the persistence tail).

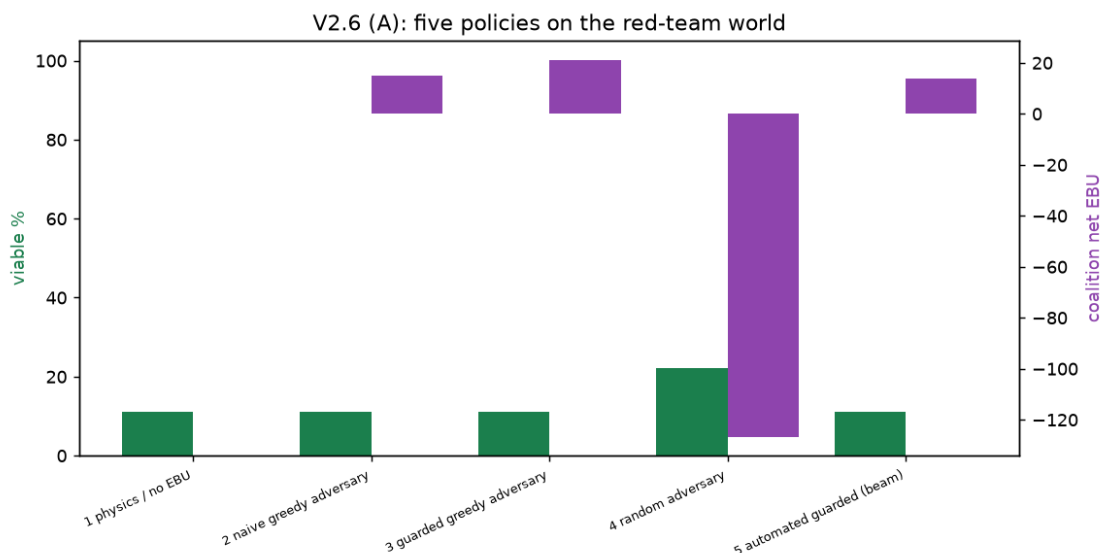


Figure 2. Five policies on the red-team fixture: viability (green) and coalition net EBU (purple). Only the random adversary worsens the field, and it loses EBU.

5. Verdict

The guarded ledger FAILED under the corrected randomized study. The hand-built red-team fixture (B) found no guarded failure and the naive positive control passed, so the search is able to expose a

known weakness; but that hand-built negative is narrow. When the properly paired randomized study (C) applied the full exploit definition — positive net EBU AND persistent physical harm vs a paired no-action baseline — it found **one confirmed profitable persistent-harm guarded trajectory** (seed 0): the adversary earns +260 EBU while driving every regenerative source to death and viability to 0%. So the corrected result overturns the earlier (mis-paired) negative: **guarded EBU is exploitable on at least one topology**.

The ledger is reported unchanged. Per the review discipline, the failing trajectory is preserved as a regression fixture and reported here; the guarded ledger was NOT modified to make the exploit disappear. Any correction belongs in a separate, clearly-labelled guarded variant, compared against this same fixture.

Mechanism / hypothesis: guarded credit is per-action and local, so a myopic greedy actor can chain locally-credited transfers whose combined multi-tick effect drives sources below the Allee threshold — each step looks locally profitable and roughly burden-neutral, but the cumulative trajectory is irreversibly worse than inaction. A lookahead objective or a persistence-aware / stock-below-reserve debit are the candidate corrections to test next.

6. Limitations and Next Steps

1. Study (C) used a 5x5 world with all-cell actors and a single greedy policy; the confirmed exploit was found by that policy, not by a full beam over (C).
2. Beam search is incomplete; a negative result (as on fixture B) reflects the searched space, not all sequences.
3. Only 12 seeds were evaluated; the true exploit rate is estimated coarsely (1/12 here).
4. Coalition size, action-menu granularity, and depth/width were kept small for runtime.

Next: (a) implement a corrected guarded variant (lookahead or persistence-aware debit) as a separate ledger and compare old-vs-corrected on the seed-0 fixture and the physical baselines, checking the fix does not reward hoarding or starve demand; (b) run the beam/red-team over the randomized layouts, not only the hand-built fixture; (c) add a learning adversary and collusion.

Reproducibility. Baseline commit, branch commit, exact commands, Python version, dependencies, seeds, and all search/experiment parameters are recorded in results/v2.6/MANIFEST.md; the captured final output is results/v2.6/v26_experiments.txt. This document is an adversarial-search study; the project has not introduced an economy.